

ANALYTIC GEOMETRY

Straight Line

- parallel → equal slopes
 $m_1 = m_2$
- perpendicular → negative reciprocals

$$m_1 = \frac{1}{m_2}$$

Equation by point-slope form:

$$m = \frac{y - y_1}{x - x_1}$$

Equation by intercept form:

$$\frac{x}{a} + \frac{y}{b} = 1$$

Equation by two-point form:

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Slope-intercept form

$$y = mx + b$$

m = slope

b = y intercept

Distance Formula

- 2 points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- point to line

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

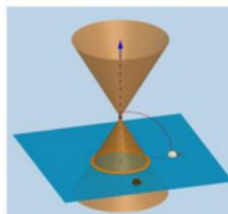
- 2 parallel lines

$$d = \frac{C_2 - C_1}{\sqrt{A^2 + B^2}}$$

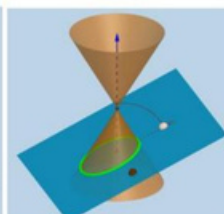
- angles between 2 lines

$$\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

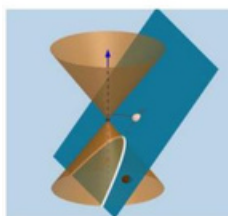
Conic Sections



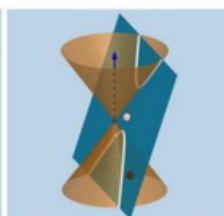
Circle



Ellipse



Parabola



Hyperbola

General Equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

Case 1: $B = 0$

*If A or C is zero, it is a **parabola**.

*If A and C are not zero and:

having opposite signs, it is a **hyperbola**.

having same sign and:

$A = C$, it is a **circle**.

$A \neq C$, it is an **ellipse**.

Case 2: $B \neq 0$

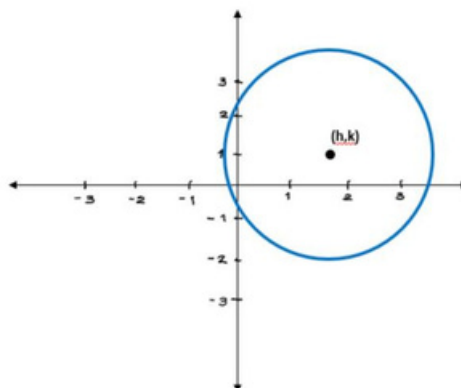
Evaluate $B^2 - 4AC$, if:

$B^2 - 4AC = 0$, it is a **parabola**.

$B^2 - 4AC < 0$, it is an **ellipse**.

$B^2 - 4AC > 0$, it is a **hyperbola**.

Circle



- General Equation

$$x^2 + y^2 + Ax + By + C = 0$$

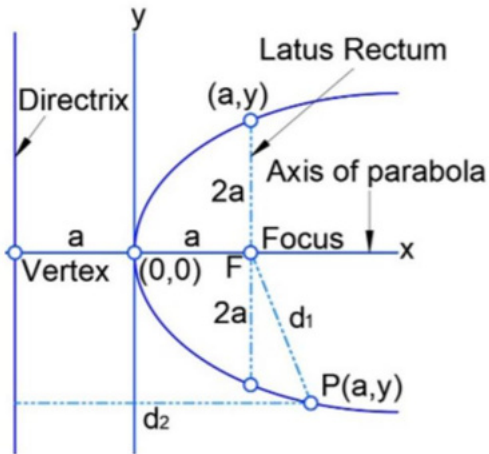
- C (0,0)

$$x^2 + y^2 = r^2$$

- C (h,k)

$$(x-h)^2 + (y-k)^2 = r^2$$

Parabola



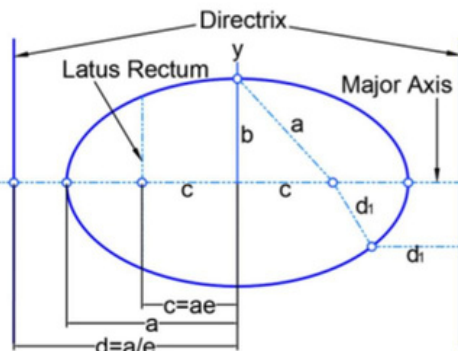
- opens upward/downward

$$(x-h)^2 = 4a(y-k)$$

- opens left/right

$$(y-k)^2 = 4a(x-h)$$

Ellipse



$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$a^2 = b^2 + c^2$$

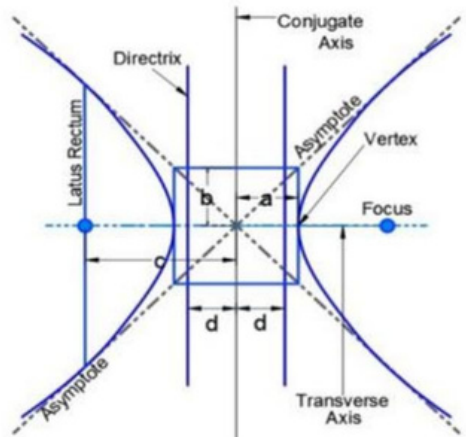
$$A = \pi ab$$

$$De = a$$

$$La = 2b^2$$

$$c = ea$$

Hyperbola



$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$a^2 + b^2 = c^2$$

- asymptote

$$y = \pm \frac{b}{a}x$$

- equilateral hyperbola

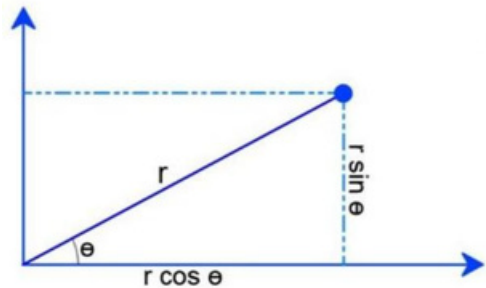
$$xy = \frac{a^2}{2}$$

$$De = a$$

$$La = 2b^2$$

$$c = ea$$

Polar Coordinate System



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r^2 = x^2 + y^2$$

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$